

William Krista Mahendra

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EDUCATION

08/2023 – 07/2024 Enschede, The Netherlands	M.Sc in Geoinformation Science and Earth Observation for Environmental Modelling and Management (GEM) Faculty of ITC, University of Twente 🌐 Thesis: <i>Assessment of Canopy Height Loss Using Sentinel-1 Time Series Data: A Case Study of Jambi Province, Indonesia</i> (Supervisors: Dr. Michael Schlund and Dr. Claudia Paris) - Analysed vegetation loss across land use and land cover classes using Sentinel-1 SAR time series, focusing on backscatter changes linked to vegetation structure. - Investigated the relationship between Sentinel-1A polarization and canopy height, and developed a Python-based automated spatio-temporal workflow for canopy height loss detection. Courses: - Radar Remote Sensing - Python Solutions
08/2022 – 06/2023 Lund, Sweden	M.Sc in Physical Geography and Ecosystem Science (GPA: VG-Pass with Distinction) Faculty of Science, Lund University 🌐 Courses: - Geographic Information Systems - Basic - Geographic Information Systems - Advanced - Ecosystem Modelling - Satellite Remote Sensing
2016 – 2020 Sleman, Indonesia	B.Sc in Geographic Information Science (GPA: 3.6/4.0 (Cum laude)) Faculty of Geography, Universitas Gadjah Mada Thesis: <i>Understanding the Influence of Different Sampling Techniques and Sample Sizes on Accuracy Assessment of Land-Use Mapping in Parts of Salatiga, Indonesia</i> (Supervisor: Prof. Projo Danoedoro)

PROFESSIONAL EXPERIENCE

08/2024 – Present Zwolle, The Netherlands	GIS and Forestry Consultant Form International - Implemented machine learning (ML) and deep learning (DL) models using multisource satellite data for land use change and deforestation mapping, achieving an overall accuracy above 90%. - Contributed to 12 carbon projects (REDD+ and ARR) across Africa, Asia, and Latin America under Verra and Plan Vivo standards, including biomass modelling, spatial analysis, and PDD writing. - Provided technical advice and knowledge transfer to project partners and clients on remote sensing analysis & GIS workflows, and carbon accounting methodologies for REDD+ and ARR projects (i.e., VM0007, VM0047). - Developed a Kalman filter based for satellite time series noise reduction workflow to improve data quality and transparency for performance benchmarking under VM0007.
01/2024 – 06/2024 Utrecht, The Netherlands	Remote Sensing Intern Acorn, Rabobank - Project Acorn supports large-scale CO₂ storage for smallholder agroforestry farmers across Africa, Asia, and Latin America. I worked under the supervision of Dr. Yaqing Gou and Dr. Elnaz Neinavaz. - Processed Sentinel-1A GRD data and evaluated multiple speckle filtering and temporal compositing aggregations, identifying Refined Lee filter combined with monthly composites as the most effective approach for reducing speckle and improving backscatter consistency for aboveground biomass (AGB) estimation. - Modelled and mapped the AGB in low biomass areas of Kigoma Region, Tanzania, using Sentinel-1 and ALOS PALSAR-2 SAR data with Random Forest and multiple regression models, achieving R² values above 0.70.
04/2021 – 05/2022 Yogyakarta, Indonesia	Research Assistant Faculty of Geography, Universitas Gadjah Mada - Participated in a joint research between University of Sydney and Universitas Gadjah Mada on analyzing Indonesian tropical peatlands. - Monitored land use/land cover change over 30 years in four peat hydrological units (>3 million ha) and assessed peat depth using integrated elevation data and field measurements. - Compiled and summarized 115 regulations for peat soil security purposes. The project result was published in Soil Security Journal.
11/2020 – 04/2021 Jakarta, Indonesia	Mapping Surveyor Directorate of Plantation, Ministry of Agriculture - Collected spatial data for more than 12,000 smallholder coconut and rubber plantations through field inventory and mobile digitization using the SW Maps platform in Purbalingga and Banten Regencies. - Delivered map and statistical results to stakeholders and partners, including area extent, productivity, and land ownership status.

PUBLICATIONS

Mahendra, W. K., Vetrira, Y., & Veronica, K. W. Integrating spaceborne LiDAR, Sentinel-1, and Sentinel-2 data with machine learning models to estimate aboveground biomass in Indonesian tropical forests. *Remote Sensing. Manuscript in preparation.*

Mahendra, W. K., & Danoedoro, P. (2024). Understanding the influence of different sample sizes and sample techniques on accuracy assessment of land cover mapping: Case study of Salatiga City, Indonesia. In *Proceedings of SPIE: Eighth Geoinformation Science Symposium: Geoinformation Science for Sustainable Planet* (Vol. 12977, Article 129770E). SPIE. <https://doi.org/10.1117/12.3009445> 

Widyatmanti, W., Minasny, B., Awanda, D., Umarhadi, D. A., Fatma, Z. S. N., **Mahendra, W. K.,** & Field, D. J. (2022). Codification to secure Indonesian peatlands: From policy to practices as revealed by remote sensing analysis. *Soil Security*, 9, 100080. <https://doi.org/10.1016/j.soisec.2022.100080> 

Danoedoro, P., Gupita, D., Afwani, M., Hadi, H., & **Mahendra, W. K.** (2022). Preliminary study on the use of digital surface models for estimating vegetation cover density in mountainous area. *Indonesian Journal of Geography*, 54(3), 333–343. <https://doi.org/10.22146/ijg.60659> 

Ambhika, N., Widyatmanti, W., **Mahendra, W. K.,** Awanda, D., & Umarhadi, D. A. (2022). Remote sensing image analysis for identification of peat thickness using spectral transformation approach: Case study of Bengkalis Island, Riau, Indonesia. *International Journal of Geoinformatics*, 18(2), 118–128. <https://doi.org/10.52939/ijg.v18i2.2161> 

Arjasakusuma, S., Kusuma, S., **Mahendra, W. K.,** & Astriviany, N. (2021). Mapping paddy field extent and temporal pattern variation in a complex terrain area using Sentinel-1 time series data: Case study of Magelang District, Indonesia. *International Journal of Geoinformatics*, 17(2), 79–88. <https://doi.org/10.52939/ijg.v17i2.1763> 

CONFERENCE PRESENTATIONS

Mahendra, W. K., & Danoedoro, P., Understanding the Influence of Different Sample Sizes and Sampling Techniques on Accuracy Assessment of Land-Cover Mapping: A Case Study of Salatiga City, Indonesia. 8th Geoinformation Science Symposium, Indonesia — Aug 2023 (**Oral Presentation**)

Mahendra, W. K., Jamaluddin, I., & Widyatmanti, W. Analysis of Image Combination Effects on Tropical Peat Swamp Mapping in Parts of Palangkaraya City, Indonesia. International Symposium on Remote Sensing (ISRS) - May 2021 (**Oral Presentation**)

Mahendra, W. K., Jamaluddin, I., & Kamal, M. Mangroves Change Detection Using Support Vector Machine Algorithm on Google Earth Engine: A Case Study in Part of the Gulf of Bone, South Sulawesi, Indonesia. 40th Asian Conference on Remote Sensing (ACRS 2019), Daejeon, South Korea — Oct 2019 (**Oral Presentation**)

Mahendra, W. K., Muharram, F. I., Alfatri, M., Muhammy, V., & Bachtiar, I. Detailed Land-Cover/Land-Use Mapping Based on SPOT-7 Imagery: A Case Study of Parepare City, Indonesia. 6th Geoinformation Science Symposium, Sleman, Indonesia — Aug 2019 (**Oral Presentation**)

PROJECTS

Project as part of professional duties at Form International:

Wikigola Project: Restoring Uganda's Forests (ARR Project) — Apr 2025 – Jan 2026

- Managed geospatial databases and conducted carbon stock calculations across four Central Forest Reserves.
- Performed time series analysis of donor pool areas using Google Earth Engine to support VM0047 compliance.

Forest Landscape Restoration in Peru's Amazon Basin — Sept 2025 – Nov 2025

- Implemented a dynamic baseline using the VM0047 methodology, achieving a standardized difference mean (SDM) of 0.17.
- Contributed to writing the Project Design Document (PDD), particularly the climate chapter.

Conkouati-Douli National REDD+ Project (CDNP), Republic of Congo (REDD+ Project) — Nov 2024 – Jan 2025

- Conducted land use change and deforestation mapping over a 10-year period across an area exceeding 250,000 ha with overall accuracy above 90%.
- Identified leakage belt areas using weighted overlay analysis to estimate future deforestation surrounding the project area.

Sustainable Forest Governance in Social Forestry Areas at Kerinci Seblat National Park, Jambi, Indonesia (CCB Project) — Aug 2024 – Jan 2025

- Conducted land use change modelling over a 10-year period to spatially map eligible REDD+ areas with overall accuracy above 90%.
- Performed carbon stock modelling for a 40-year project period to support REDD+ calculations.
- Managed geospatial databases to ensure compliance with auditing requirements.

Full projects are available through this website: <https://sites.google.com/view/wmahendra/projects> 

Research Project as part of Studies

Department of Natural Resource and Management, Faculty of ITC, University of Twente:

- Deforestation detection in the Brazilian Amazon forest using BFAST algorithm and MODIS time series data, achieving overall accuracy of 82%. - Dec 2023 - Feb 2024
- Assessment of polarimetric Synthetic Aperture Radar using dual-polarization Sentinel-1A data for land use and land cover mapping in a complex landscape of Bengkalis Island, Indonesia. - Sept 2023 - Nov 2023

Department of Physical Geography and Ecosystem Science, Lund University:

- Monitored Vegetation Phenology in Northern Latitude of Norway retrieved by MODIS time series data - May 2023 - Jun 2023
- Soil Hydrology Modelling: case study of Hyytiälä, Finland - Mar - May 2023
- The impact of extreme temperature on Carbon Flux (i.e., GPP and NPP) using Landsat-8 OLI in Scania Region, Sweden - Dec 2022 - Feb 2023

Research project Department of Geographic Information Science, Universitas Gadjah Mada:

- Land cover and land use Mapping in Salatiga City, Indonesia - Nov 2019 - Aug 2020
- Vegetation carbon mapping in mountainous areas of Parepare city, Indonesia - Dec 2018 - Jun 2019

TEACHING AND ADVISING

Practicum Assistant for Undergraduate Students at the Remote Sensing Laboratory, Faculty of Geography, Universitas Gadjah Mada

Remote Sensing for Landscape Ecology - Aug-Dec 2019

- Instructed field based LULC surveys and vegetation biomass measurements, and their integration of field data with satellite datasets.

Digital Image Processing for Satellite Data - Jan-Jun 2019

- Taught atmospheric correction, geometric correction, principal component analysis (PCA), and supervised/unsupervised image classification using various algorithms.

Remote Sensing Non-Photography System - Jan-Jun 2019

- Taught satellite system principles, electromagnetic wavelength characteristics, and spectral responses of different land-cover types using ENVI software.

Remote Sensing Photography System - Aug-Dec 2018

- Taught fundamental concepts of drone operation, flight planning, GCP/ICP selection, image mosaicking and orthophoto analyses using Agisoft PhotoScan.

SKILLS

Geospatial Data Processing

- Proficient in using Python and R to analyse geospatial raster data.
- Python: Rasterio, GeoPandas, GEEMap, Matplotlib, TensorFlow, FilterPy, Seaborn
- R : raster, lidR, TreeLS, sf, ggplot2, rgeos, randomForest, MatchIt
- Proficient in using image processing software, such as SNAP, PolSAR Pro, ENVI and IDRISI.
- Proficient in using Fusion and Agisoft Metashape for LiDAR processing and visualization.

Google Earth Engine

- Competent in data collection, image classification, regression, and Time Series Analysis.

Time Series and Ecosystem Modelling

Intermediate proficiency in TIMESAT and LPJ-GUESS

- Satellite time-series analysis and dynamic vegetation and ecosystem modelling.

Data Analysis and Visualization

Proficient in using QGIS, ArcGIS Products, and Microsoft Products (Microsoft Power Point and Microsoft Excel).

Written and Verbal Communication

- As a young researcher, I gained good public speaking and knowledge transfer from experiences in presenting in four different international conferences.
- I also engaged with people of all levels from gathering, work and presenting solutions.

INTEREST KEYWORDS

Optical and SAR Remote Sensing, Light Detection and Ranging (LiDAR), GIS, Time Series, Landscape Ecology, Land Use and Land Cover (LULC), Image Processing, Machine Learning, Deep Learning, Unmanned Aerial Vehicles (UAVs), Forest Carbon Modelling, Nature Based Solutions, Carbon Credits, REDD+, Afforestation, Reforestation and Revegetation (ARR), Spatial Statistics, Soil analysis, Conservation.

HONORS AND FELLOWSHIP

20 Aug 2023

2nd Place – Holland Busathon Competition University of Twente & Vrije Universiteit Amsterdam

- Addressed delayed tulip harvests in Den Helder, the Netherlands by integrating geospatial and economic approaches, applying spatial analysis and weather-based assessments to predict optimal harvest periods.
- Presented findings to GIS specialists, business practitioners, and academic staff to support decision-making and inform further research.

Jun 2022

Erasmus Mundus Joint Master Scholarship Recipient European Commission

- Selected as 1 of 15 full scholarship recipients from over 550 global applicants to pursue a master's degree (€ 49,000).
- Enrolled in the prestigious GEM Programme (Geoinformation Science and Earth Observation for Environmental Modelling and Monitoring).

27 Aug 2021

Best Conference Paper Award The 7th Geoinformation Science Symposium

- Awarded for the paper titled: "Deep Neural Network Regression for Estimating Land Surface Temperature at 10-Meter Spatial Resolution Using Landsat-8 and Sentinel-2 Data."

29 Jun 2019

Community Development Grant Award Ministry of Villages, Development of Disadvantaged Regions, and Transmigration (Indonesia)

- Received a grant to support community development initiatives aimed at reducing child stunting in Ponu Village by conducting integrated healthcare activities, supporting stakeholder education, and developing a geospatial database to assist with analysis and planning (IDR 60,000,000).

LANGUAGES

English - C1 (Fluent)

Indonesian - Native

Dutch - A2 (Conversational)

Swedish - Basic

PROFESSIONAL MEMBERSHIPS

2025 – Present

European Geosciences Union (Earth and Space Science division)

05/2025 – Present

Indonesian Professional Association (IPA), The Netherlands

01/2025 – Present

Gadjah Mada University Alumni Association in the Netherlands (Career Network and Media Staff)

11/2020 – Present

Indonesian Geographer Association (Member)